(P) **7** The displacement of a particle at time t is given by the function $f(t)$, where $f(0) = 0$.
Given that the velocity of the particle is given by $f'(t) = 10 - 5t$,

a find $f(t)$

b determine the displacement of the particle when $t = 3$.

Problem-solving

You don't need any specific knowledge of mechanics to answer this question. You are told that the displacement of the particle at time t is given by $f(t)$.

(P) **8** The height, in metres, of an arrow fired horizontally from the top of a castle is modelled by the function $f(t)$, where $f(0) = 35$. Given that $f'(t) = -9.8t$,

a find $f(t)$.

b determine the height of the arrow when $t = 1.5$.

c write down the height of the castle according to this model.

d estimate the time it will take the arrow to hit the ground.

e state one assumption used in your calculation.

Challenge

1 A set of curves, where each curve passes through the origin, has equations $y = f_1(x)$, $y = f_2(x)$, $y = f_3(x)$ … where $f_n'(x) = f_{n-1}(x)$ and $f_1(x) = x^2$.

a Find $f_2(x)$, $f_3(x)$.

b Suggest an expression for $f_n(x)$.

2 A set of curves, with equations $y = f_1(x)$, $y = f_2(x)$, $y = f_3(x)$, … all pass through the point (0, 1) and they are related by the property $f_n'(x) = f_{n-1}(x)$ and $f_1(x) = 1$. Find $f_2(x)$, $f_3(x)$, $f_4(x)$.

13.4 Definite integrals

You can calculate an integral between two **limits**. This is called a **definite integral**. A definite integral usually produces a **value** whereas an indefinite integral always produces a **function**.

Here are the steps for integrating the function $3x^2$ between the limits $x = 1$ and $x = 2$.

The limits of the integral are from $x = 1$ to $x = 2$.

$$\left[\int_1^2 3x^2\,dx = [x^3]_1^2\right]$$

Write the integral in [ ] brackets.

$$= (2^3) - (1^3)$$

Write this step in ( ) brackets.

Evaluate the integral at the upper limit.

$$= 8 - 1$$

$$= 7$$

Evaluate the integral at the lower limit.

There are three stages when you work out a definite integral:

Write the definite integral **statement** with its limits, a and b.	**Integrate**, and write the integral in square brackets	**Evaluate** the definite integral by working out $f(b) - f(a)$.
$\int_a^b \ldots dx$	$[\ldots]_a^b$	$(\ldots) - (\ldots)$

- **If $f'(x)$ is the derivative of $f(x)$ for all values of x in the interval $[a, b]$, then the definite integral is defined as**
  $\int_a^b f'(x)dx = [f(x)]_a^b = f(b) - f(a)$.

Problem-solving

The relationship between the derivative and the integral is called the **fundamental theorem of calculus**.

Example 7

Evaluate

$\int_0^1 (x^{\frac{1}{3}} - 1)^2 dx$

$\int_0^1 (x^{\frac{1}{3}} - 1)^2 dx$

$= \int_0^1 (x^{\frac{2}{3}} - 2x^{\frac{1}{3}} + 1)dx$

First multiply out the bracket to put the expression in a form ready to be integrated.

$= \left[\frac{x^{\frac{5}{3}}}{\frac{5}{3}} - 2\frac{x^{\frac{4}{3}}}{\frac{4}{3}} + x\right]_0^1$

For definite integrals you don't need to include $+c$ in your square brackets.

$= \left[\frac{3}{5}x^{\frac{5}{3}} - \frac{3}{2}x^{\frac{4}{3}} + x\right]_0^1$

Simplify each term.

$= \left(\frac{3}{5} - \frac{3}{2} + 1\right) - (0 + 0 + 0)$

$= \frac{1}{10}$

Example 8

Given that P is a constant and $\int_1^5 (2Px + 7)dx = 4P^2$, show that there are two possible values for P and find these values.

$\int_1^5 (2Px + 7)dx = [Px^2 + 7x]_1^5$

$= (25P + 35) - (P + 7)$

$= 24P + 28$

$24P + 28 = 4P^2$

$4P^2 - 24P - 28 = 0$

$P^2 - 6P - 7 = 0$

Divide every term by 4 to simplify.

$(P + 1)(P - 7) = 0$

$P = -1$ or 7

Problem-solving

You are integrating with respect to x so treat P as a constant. Find the definite integral in terms of P then it equal to $4P^2$. The fact that the question asks for 'two possible values' gives you a clue that the resulting equation will be quadratic.

Exercise 13D

1 Evaluate the following definite integrals:

a $\int_2^5 x^3\,dx$ **b** $\int_1^3 x^4\,dx$

c $\int_0^4 \sqrt{x}\,dx$ **d** $\int_1^3 \frac{3}{x^2}\,dx$

Watch out You must not use a calculator to work out definite integrals in your exam. You need to use calculus and show clear algebraic working.

2 Evaluate the following definite integrals:

a $\int_1^2 \left(\frac{2}{x^3} + 3x\right)dx$ **b** $\int_0^2 (2x^3 - 4x + 5)dx$ **c** $\int_4^9 \left(\sqrt{x} - \frac{6}{x^2}\right)dx$ **d** $\int_1^8 (x^{-\frac{1}{3}} + 2x - 1)dx$

3 Evaluate the following definite integrals:

a $\int_1^3 \frac{x^3 + 2x^2}{x}\,dx$ **b** $\int_3^6 \left(x - \frac{3}{x}\right)^2 dx$ **c** $\int_0^1 x^2\left(\sqrt{x} + \frac{1}{x}\right)dx$ **d** $\int_1^4 \frac{2 + \sqrt{x}}{x^2}\,dx$

E/P **4** Given that A is a constant and $\int_1^4 (6\sqrt{x} - A)dx = A^2$, show that there are two possible values for A and find these values. **(5 marks)**

E **5** Use calculus to find the value of $\int_1^9 (2x - 3\sqrt{x})dx$. **(5 marks)**

E **6** Evaluate $\int_4^{12} \frac{2}{\sqrt{x}}\,dx$, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers. **(4 marks)**

E/P **7** Given that $\int_1^k \frac{1}{\sqrt{x}}\,dx = 3$, calculate the value of k. **(4 marks)**

Problem-solving You might encounter a definite integral with an unknown in the limits. Here, you can find an expression for the definite integral in terms of k then set that expression equal to 3.

8 The speed, v ms^{-1}, of a train at time t seconds is given by $v = 20 + 5t$, $0 \leqslant t \leqslant 10$.

The distance, s metres, travelled by the train in 10 seconds is given by $s = \int_0^{10} (20 + 5t)dt$. Find the value of s.

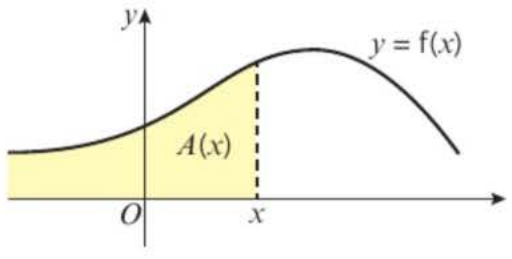
Challenge

Given that $\int_k^{3k} \frac{3x + 2}{8}\,dx = 7$ and $k > 0$, calculate the value of k.

13.5 Areas under curves

Definite integration can be used to find the area under a curve.

For any curve with equation $y = f(x)$, you can define the area under the curve to the left of x as a function of x called $A(x)$. As x increases, this area $A(x)$ also increases (since x moves further to the right).

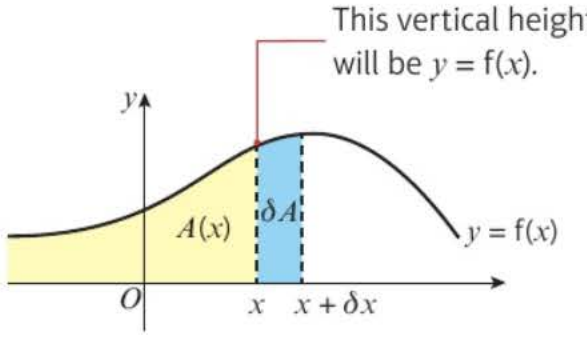


If you look at a small increase in x, say δx, then the area increases by an amount $\delta A = A(x + \delta x) - A(x)$.

This increase in the δA is approximately rectangular and of magnitude $y\delta x$. (As you make δx smaller any error between the actual area and this will be negligible.)

So you have $\delta A \approx y\delta x$

or $\dfrac{\delta A}{\delta x} \approx y$



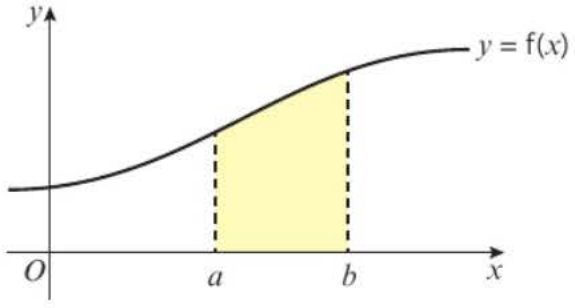
and if you take the limit $\lim_{\delta x \to 0}\left(\dfrac{\delta A}{\delta x}\right)$ then you will see that $\dfrac{dA}{dx} = y$.

Now if you know that $\dfrac{dA}{dx} = y$, then to find A you have to integrate, giving $A = \int y\,dx$.

- **The area between a positive curve, the x-axis and the lines $x = a$ and $x = b$ is given by**

  Area $= \int_a^b y\,dx$

  where $y = f(x)$ is the equation of the curve.

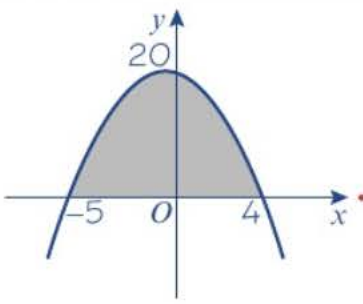


Example 9

Find the area of the finite region between the curve with equation $y = 20 - x - x^2$ and the x-axis.

$y = 20 - x - x^2 = (4 - x)(5 + x)$

Factorise the expression.



Draw a sketch of the graph. $x = 4$ and $x = -5$ are the points of intersection of the curve and the x-axis.

$$\text{Area} = \int_{-5}^{4}(20 - x - x^2)dx$$

$$= \left[20x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-5}^{4}$$

$$= (80 - 8 - \frac{64}{3}) - (-100 - \frac{25}{2} + \frac{125}{3})$$

$$= \frac{243}{2}$$

You don't normally need to give units when you are finding areas on graphs.

Exercise 13E

1 Find the area between the curve with equation $y = f(x)$, the x-axis and the lines $x = a$ and $x = b$ in each of the following cases:

a $f(x) = -3x^2 + 17x - 10$; $a = 1$, $b = 3$

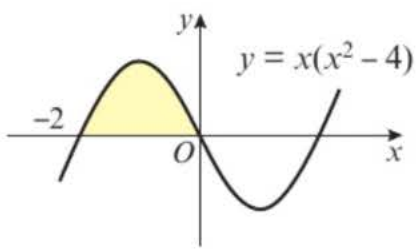
b $f(x) = 2x^3 + 7x^2 - 4x$; $a = -3$, $b = -1$

c $f(x) = -x^4 + 7x^3 - 11x^2 + 5x$; $a = 0$, $b = 4$

d $f(x) = \dfrac{8}{x^2}$; $a = -4$, $b = -1$

Hint For part **c**, $f(x) = -x(x - 1)^2(x - 5)$

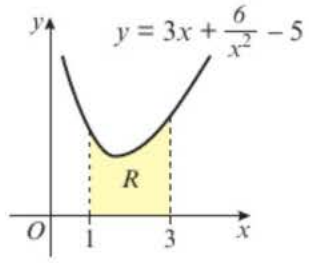
2 The sketch shows part of the curve with equation $y = x(x^2 - 4)$. Find the area of the shaded region.



3 The diagram shows a sketch of the curve with equation $y = 3x + \dfrac{6}{x^2} - 5$, $x > 0$.

The region R is bounded by the curve, the x-axis and the lines $x = 1$ and $x = 3$.

Find the area of R.

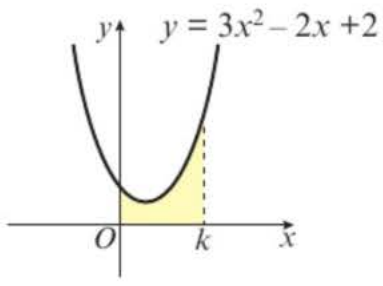


4 Find the area of the finite region between the curve with equation $y = (3 - x)(1 + x)$ and the x-axis.

5 Find the area of the finite region between the curve with equation $y = x(x - 4)^2$ and the x-axis.

(P) **6** Find the area of the finite region between the curve with equation $y = 2x^2 - 3x^3$ and the x-axis.

(P) **7** The shaded area under the graph of the function $f(x) = 3x^2 - 2x + 2$, bounded by the curve, the x-axis and the lines $x = 0$ and $x = k$, is 8. Work out the value of k.



Problem-solving

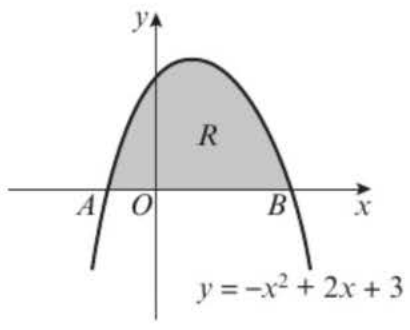
$\int_0^k (3x^2 - 2x + 2)\,dx = 8$

(E) **8** The finite region R is bounded by the x-axis and the curve with equation $y = -x^2 + 2x + 3$, $x \geqslant 0$.

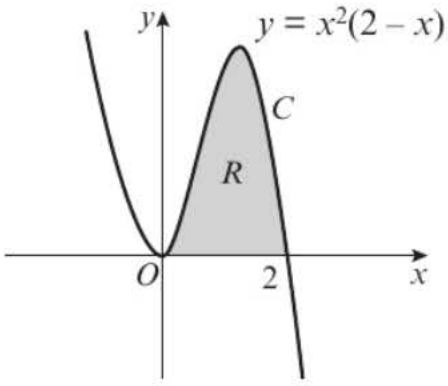
The curve meets the x-axis at points A and B.

a Find the coordinates of point A and point B. **(2 marks)**

b Find the area of the region R. **(4 marks)**



E **9** The graph shows part of the curve C with equation $y = x^2(2 - x)$.
The region R, shown shaded, is bounded by C and the x-axis.
Use calculus to find the exact area of R. **(5 marks)**



Watch out If a question says "use calculus" then you need to use integration or differentiation, and show clear algebraic working.

13.6 Areas under the x-axis

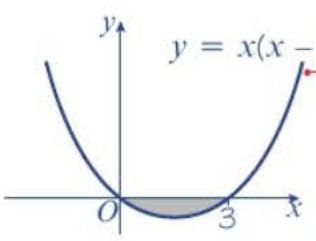
You need to be careful when you are finding areas below the x-axis.

- **When the area bounded by a curve and the x-axis is below the x-axis, $\int y\,dx$ gives a negative answer.**

Example 10

Find the area of the finite region bounded by the curve $y = x(x - 3)$ and the x-axis.

When $x = 0$, $y = 0$
When $y = 0$, $x = 0$ or 3



First sketch the curve.
It is ⋁-shaped and crosses the x-axis at 0 and 3.

$\text{Area} = \int_0^3 x(x - 3)dx$

The limits on the integral will therefore be 0 and 3.

$= \int_0^3 (x^2 - 3x)dx$

Multiply out the brackets.

$= \left[\frac{x^3}{3} - \frac{3x^2}{2}\right]_0^3$

Integrate as usual.

$= \left(\frac{27}{3} - \frac{27}{2}\right) - (0 - 0)$

$= -\frac{27}{6}$ or $-\frac{9}{2}$ or -4.5

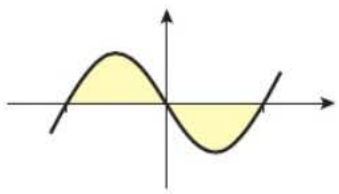
The area is below the x-axis so the definite integral is negative.

So the area is 4.5

State the area as a positive quantity.

Online Check your solution using your calculator.

The following example shows that great care must be taken if you are trying to find an area which straddles the x-axis such as the shaded region.



For examples of this type you need to draw a sketch, unless one is given in the question.

Example 11

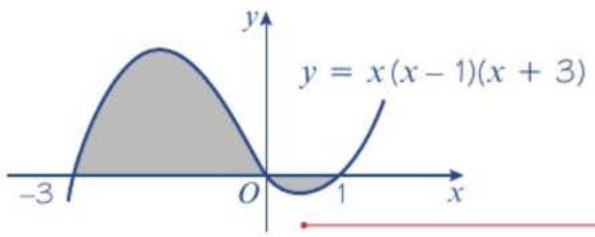
Sketch the curve with equation $y = x(x - 1)(x + 3)$ and find the area of the finite region bounded by the curve and the x-axis.

When $x = 0$, $y = 0$
When $y = 0$, $x = 0$, 1 or -3

Find out where the curve cuts the axes.

$x \to \infty$, $y \to \infty$
$x \to -\infty$, $y \to -\infty$

Find out what happens to y when x is large and positive or large and negative.



Problem-solving

Always draw a sketch, and use the points of intersection with the x-axis as the limits for your integrals.

The area is given by $\int_{-3}^{0} y\,dx - \int_{0}^{1} y\,dx$

Now $\int y\,dx = \int (x^3 + 2x^2 - 3x)dx$

$$= \left[\frac{x^4}{4} + \frac{2x^3}{3} - \frac{3x^2}{2}\right]$$

Since the area between $x = 0$ and 1 is below the axis the integral between these points will give a negative answer.

So $\int_{-3}^{0} y\,dx = (0) - \left(\frac{81}{4} - \frac{2}{3} \times 27 - \frac{3}{2} \times 9\right)$

Multiply out the brackets.

$$= \frac{45}{4}$$

and $\int_{0}^{1} y\,dx = \left(\frac{1}{4} + \frac{2}{3} - \frac{3}{2}\right) - (0)$

$$= -\frac{7}{12}$$

So the area required is $\frac{45}{4} + \frac{7}{12} = \frac{71}{6}$

Watch out If you try to calculate the area as a single definite integral, the positive and negative areas will partly cancel each other out.

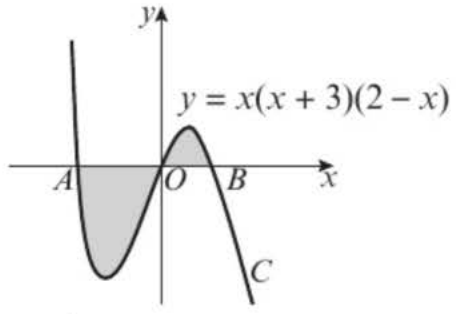
Exercise 13F

1 Sketch the following and find the total area of the finite region or regions bounded by the curves and the x-axis:

a $y = x(x + 2)$ **b** $y = (x + 1)(x - 4)$ **c** $y = (x + 3)x(x - 3)$

d $y = x^2(x - 2)$ **e** $y = x(x - 2)(x - 5)$

E **2** The graph shows a sketch of part of the curve C with equation $y = x(x + 3)(2 - x)$.



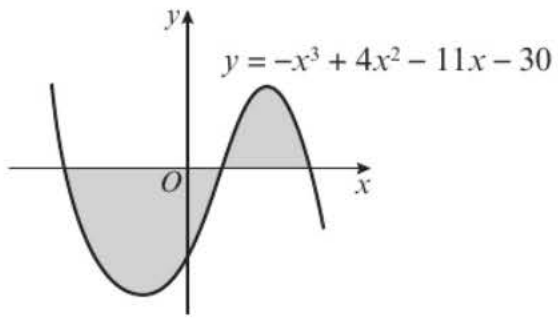
The curve C crosses the x-axis at the origin O and at points A and B.

a Write down the x-coordinates of A and B. **(1 mark)**

The finite region, shown shaded, is bounded by the curve C and the x-axis.

b Use integration to find the total area of the finite shaded region. **(7 marks)**

3 $f(x) = -x^3 + 4x^2 + 11x - 30$



The graph shows a sketch of part of the curve with equation $y = -x^3 + 4x^2 + 11x - 30$.

a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

b Write $f(x)$ in the form $(x + 3)(Ax^2 + Bx + C)$.

c Hence, factorise $f(x)$ completely.

d Hence, determine the x-coordinates where the curve intersects the x-axis.

e Hence, determine the total shaded area shown on the sketch.

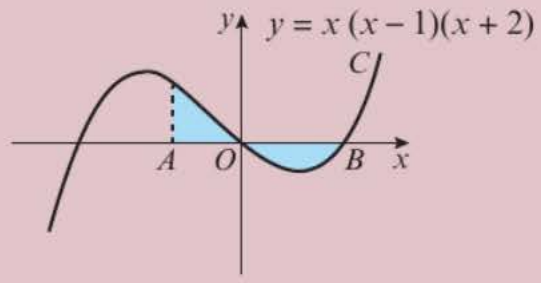
Challenge

1 Given that $f(x) = x(3 - x)$, find the area of the finite region bounded by the x-axis and the curve with equation

a $y = f(x)$ **b** $y = 2f(x)$ **c** $y = af(x)$

d $y = f(x + a)$ **e** $y = f(ax)$.

2 The graph shows a sketch of part of the curve C with equation $y = x(x - 1)(x + 2)$. The curve C crosses the x-axis at the origin O and at point B.
The shaded areas above and below the x-axis are equal.



a Show that the x-coordinate of A satisfies the equation

$$(x - 1)^2(3x^2 + 10x + 5) = 0$$

b Hence find the exact coordinates of A, and interpret geometrically the other roots of this equation.

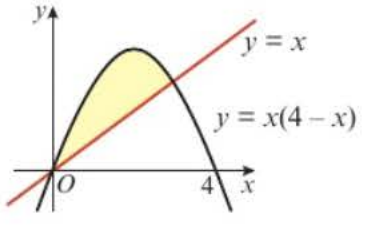
13.7 Areas between curves and lines

- **You can use definite integration together with areas of trapeziums and triangles to find more complicated areas on graphs.**

Example 12

The diagram shows a sketch of part of the curve with equation $y = x(4 - x)$ and the line with equation $y = x$.

Find the area of the region bounded by the curve and the line.



$x(4 - x) = x$
$3x - x^2 = 0$
$x(3 - x) = 0$
$x = 0$ or 3

Area beneath curve $= \int_0^3 (4x - x^2)dx$

$= \left[2x^2 - \frac{x^3}{3}\right]_0^3$

$= 9$

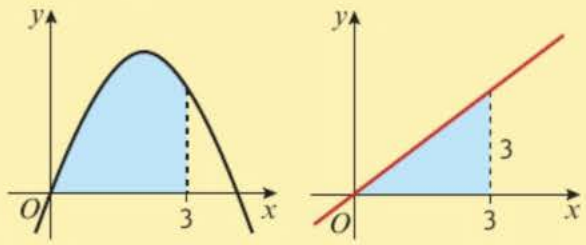
Area beneath triangle $= \frac{1}{2} \times 3 \times 3$

$= \frac{9}{2}$

Shaded area $= 9 - \frac{9}{2} = \frac{9}{2}$

First, find the x-coordinate of the points of intersection of the curve $y = x(4 - x)$ and the line $y = x$.

Shaded area = area beneath curve – area beneath triangle

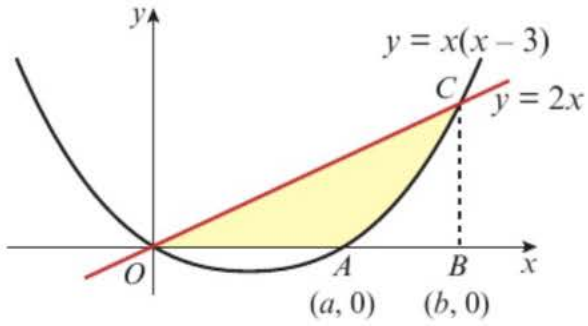


$\left[2x^2 - \frac{x^3}{3}\right]_0^3 = \left(18 - \frac{27}{3}\right) - (0 - 0) = 18 - 9$

Example 13

The diagram shows a sketch of the curve with equation $y = x(x - 3)$ and the line with equation $y = 2x$.

Find the area of the shaded region OAC.



The required area is given by:

Area of triangle $OBC - \int_a^b x(x - 3)dx$

The curve cuts the x-axis at $x = 3$ (and $x = 0$) so $a = 3$.

The curve meets the line $y = 2x$ when $2x = x(x - 3)$.

So $0 = x^2 - 5x$

$0 = x(x - 5)$

$x = 0$ or 5, so $b = 5$

The point C is $(5, 10)$.

Area of triangle $OBC = \frac{1}{2} \times 5 \times 10 = 25$.

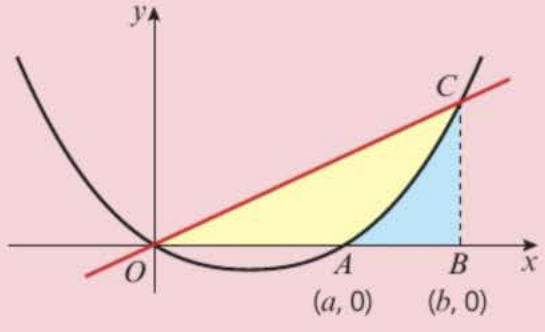
Area between curve, x-axis and the line $x = 5$ is

$\int_3^5 x(x - 3)dx = \int_3^5 (x^2 - 3x)dx$

$= \left[\frac{x^3}{3} - \frac{3x^2}{2}\right]_3^5$

Problem-solving

Look for ways of combining triangles, trapeziums and direct integrals to find the missing area.



Substituting $x = 5$ into the equation of the line gives $y = 2 \times 5 = 10$.

Work out the definite integral separately. This will help you avoid making errors in your working.

$$= \left(\frac{125}{3} - \frac{75}{2}\right) - \left(\frac{27}{3} - \frac{27}{2}\right)$$

$$= \left(\frac{25}{6}\right) - \left(-\frac{27}{6}\right)$$

$$= \frac{26}{3}$$

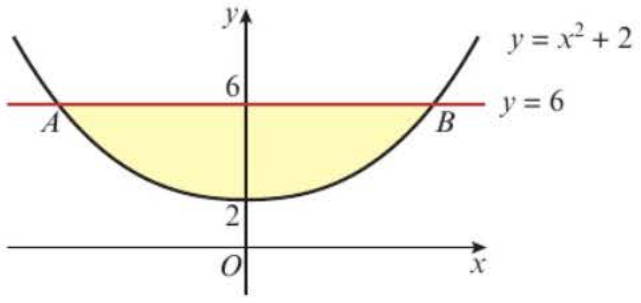
Shaded region is therefore $= 25 - \frac{26}{3} = \frac{49}{3}$

Exercise 13G

1 The diagram shows part of the curve with equation $y = x^2 + 2$ and the line with equation $y = 6$.
The line cuts the curve at the points A and B.

a Find the coordinates of the points A and B.

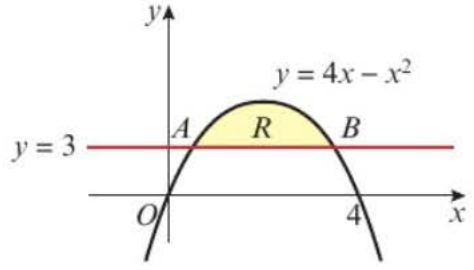
b Find the area of the finite region bounded by line AB and the curve.



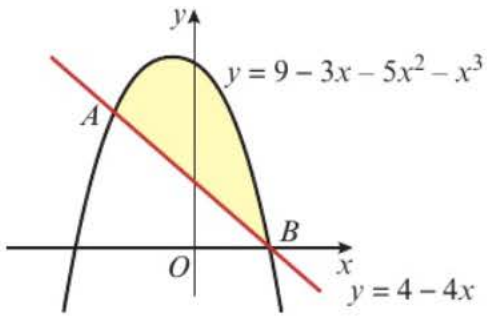
2 The diagram shows the finite region, R, bounded by the curve with equation $y = 4x - x^2$ and the line $y = 3$.
The line cuts the curve at the points A and B.

a Find the coordinates of the points A and B.

b Find the area of R.



(P) **3** The diagram shows a sketch of part of the curve with equation $y = 9 - 3x - 5x^2 - x^3$ and the line with equation $y = 4 - 4x$.
The line cuts the curve at the points $A(-1, 8)$ and $B(1, 0)$.
Find the area of the shaded region between AB and the curve.

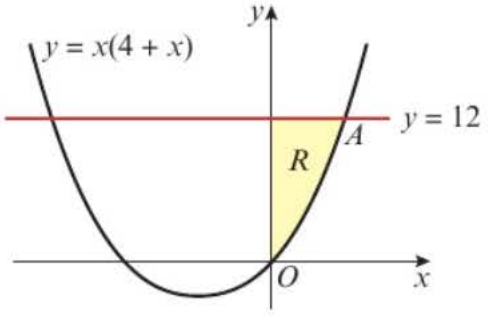


(P) **4** Find the area of the finite region bounded by the curve with equation $y = (1 - x)(x + 3)$ and the line $y = x + 3$.

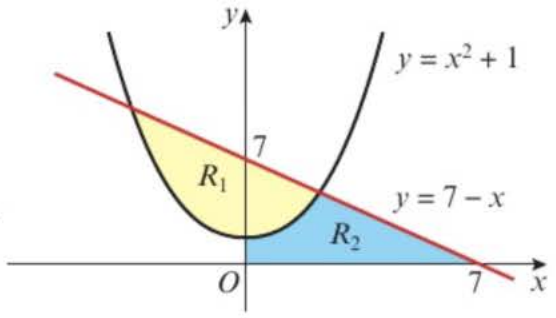
5 The diagram shows the finite region, R, bounded by the curve with equation $y = x(4 + x)$, the line with equation $y = 12$ and the y-axis.

a Find the coordinates of the point A where the line meets the curve.

b Find the area of R.



P **6** The diagram shows a sketch of part of the curve with equation $y = x^2 + 1$ and the line with equation $y = 7 - x$.
The finite region, R_1 is bounded by the line and the curve.
The finite region, R_2 is below the curve and the line and is bounded by the positive x- and y-axes as shown in the diagram.



a Find the area of R_1.

b Find the area of R_2.

P **7** The curve C has equation $y = x^{\frac{2}{3}} - \dfrac{2}{x^{\frac{1}{3}}} + 1$.

a Verify that C crosses the x-axis at the point $(1, 0)$.

b Show that the point $A(8, 4)$ also lies on C.

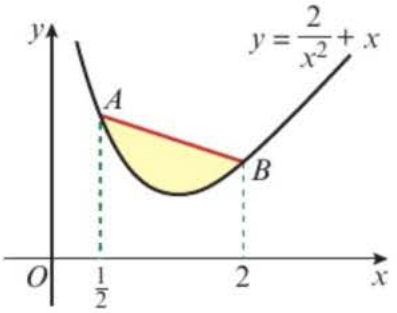
c The point B is $(4, 0)$. Find the equation of the line through AB.

The finite region R is bounded by C, AB and the positive x-axis.

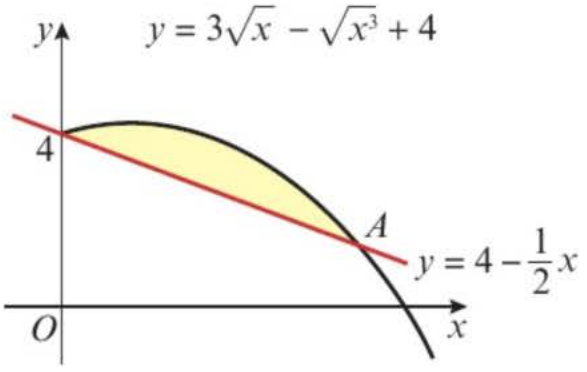
d Find the area of R.

P **8** The diagram shows part of a sketch of the curve with equation $y = \dfrac{2}{x^2} + x$. The points A and B have x-coordinates $\frac{1}{2}$ and 2 respectively.

Find the area of the finite region between AB and the curve.



P **9** The diagram shows part of the curve with equation $y = 3\sqrt{x} - \sqrt{x^3} + 4$ and the line with equation $y = 4 - \frac{1}{2}x$.

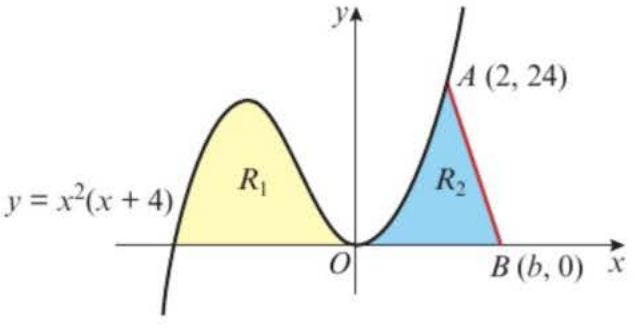


a Verify that the line and the curve cross at the point $A(4, 2)$.

b Find the area of the finite region bounded by the curve and the line.

P **10** The sketch shows part of the curve with equation $y = x^2(x + 4)$. The finite region R_1 is bounded by the curve and the negative x-axis. The finite region R_2 is bounded by the curve, the positive x-axis and AB, where $A(2, 24)$ and $B(b, 0)$.

The area of R_1 = the area of R_2.



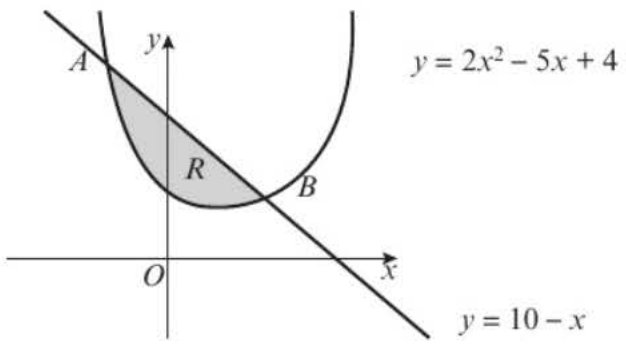
a Find the area of R_1.

b Find the value of b.

Problem-solving

Split R_2 into two areas by drawing a vertical line at $x = 2$.

E/P **11** The line with equation $y = 10 - x$ cuts the curve with equation $y = 2x^2 - 5x + 4$ at the points A and B, as shown.



a Find the coordinates of A and the coordinates of B. **(5 marks)**

The shaded region R is bounded by the line and the curve as shown.

b Find the exact area of R. **(6 marks)**

Mixed exercise 13

1 Find:

a $\int (x + 1)(2x - 5)\mathrm{d}x$ **b** $\int (x^{\frac{1}{3}} + x^{-\frac{1}{3}})\mathrm{d}x$

2 The gradient of a curve is given by $\mathrm{f}'(x) = x^2 - 3x - \frac{2}{x^2}$. Given that the curve passes through the point (1, 1), find the equation of the curve in the form $y = \mathrm{f}(x)$.

3 Find:

a $\int (8x^3 - 6x^2 + 5)\mathrm{d}x$ **b** $\int (5x + 2)x^{\frac{1}{2}}\mathrm{d}x$

P **4** Given $y = \frac{(x + 1)(2x - 3)}{\sqrt{x}}$, find $\int y\,\mathrm{d}x$.

P **5** Given that $\frac{\mathrm{d}x}{\mathrm{d}t} = (t + 1)^2$ and that $x = 0$ when $t = 2$, find the value of x when $t = 3$.

E/P **6** Given that $y^{\frac{1}{2}} = x^{\frac{1}{3}} + 3$:

a show that $y = x^{\frac{2}{3}} + Ax^{\frac{1}{3}} + B$, where A and B are constants to be found. **(2 marks)**

b hence find $\int y\,\mathrm{d}x$. **(3 marks)**

E/P **7** Given that $y^{\frac{1}{2}} = 3x^{\frac{1}{4}} - 4x^{-\frac{1}{4}}$ $(x > 0)$:

a find $\frac{\mathrm{d}y}{\mathrm{d}x}$ **(2 marks)**

b find $\int y\,\mathrm{d}x$. **(3 marks)**

P **8** $\int \left(\frac{a}{3x^3} - ab\right)\mathrm{d}x = -\frac{2}{3x^2} + 14x + c$

Find the value of a and the value of b.

P **9** A rock is dropped off a cliff. The height in metres of the rock above the ground after t seconds is given by the function $\mathrm{f}(t)$. Given that $\mathrm{f}(0) = 70$ and $\mathrm{f}'(t) = -9.8t$, find the height of the rock above the ground after 3 seconds.